Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
8	*(a)*	(i)	No emission if photon energy < ϕ [or emission only if photon energy > ϕ](1) Convincing argument, clearly implying that photon energy = hf and leading to no emission if $f < \frac{\phi}{h}$ (1) Increasing light intensity just gives more photons **or** doesn't change [energy of] individual photons **or** doesn't help because photons don't co-operate (1)	3			3		
		(ii)	Photon energy = 4.37×10^{-19} [J] (1) No emission by Ca or Zn (or equiv) (1) $KE_{\text{max}} < 0.56 \times 10^{-19}$ J or $\phi > 3.81 \times 10^{-19}$ J or equivalent, e.g. V_S = 0.78 V (Cs), 0.43 V (K), 0.21 V (Ba) (1) Therefore Ba (1) [award mark only if attempted justification] See below:			4	4	2	
	(b)	(i)	Acceptable straight line through points (going through origin would be unacceptable)		1		1	1	1
		(ii)	Straight line as predicted (1) But not through origin (or non-zero intercept). Equation predicts through origin (or proportionality) (1)			2	2	1	2
		(iii)	Data from graph put into $\frac{\Delta V}{\Delta f}$ irrespective of slips such as incorrect powers of 10 (1) Accept between 3.90×10^{-15} and 4.40×10^{-15} [V Hz^{-1}] and to either 2 or 3 sig figs (1) $h = 6.64 \times 10^{-34}$ J s **ecf** from gradient (1)			3	3	2	3
			Question 8 total	**3**	**1**	**9**	**13**	**6**	**6**